第零章　預備知識

0.1 基本概念

記號

$\forall$	對所有	for all
$\exists$	存在	there exists
$\exists!$	存在唯一	there exists uniquely
$\in$	屬於	belongs to
A $\Longrightarrow$ B	若 A 則 B	if A then B
A $\Longleftrightarrow$ B	A 等價於 B	A if and only if B
∞	無限大	infinity
$\vee$	或	or
$\wedge$	且	and
$\because$	因為	because
$\therefore$	所以	therefore
$\neg$	非	not
$\equiv$	等價於	is equivalent to

數

$\mathbb{N}$	自然數	natural number	$1, 2, 3, \ldots$
$\mathbb{Z}$	整數	integer	$\ldots, -2, -1, 0, 1, 2, \ldots$
$\mathbb{Q}$	有理數	rational number	$\dfrac{p}{q} : p, q \in \mathbb{Z}$
$\mathbb{R}$	實數	real number	
$\mathbb{C}$	複數	complex number	$\alpha + \beta\,\mathrm{i} : \alpha, \beta \in \mathbb{R}, \mathrm{i} = \sqrt{-1}$

集合

$x \in S$	x 為集合 S 的元素	
$S_1 = \{x_1, x_2, \ldots\}$	列舉式	
$S_2 = \{x \mid x$ 滿足某性質$\}$	敘述式	
$S \cap T$	$\{x \mid x \in S \wedge x \in T\}$	交集 (intersection)
$S \cup T$	$\{x \mid x \in S \vee x \in T\}$	聯集 (union)
$S \setminus T$	$\{x \mid x \in S \wedge x \notin T\}$	差集 (difference)
$S \times T$	$\{(x, y) \mid x \in S \wedge y \in T\}$	積集 (Cartesian product)
$\varnothing$	空集合	
$S_1 \subset S_2$, $S_2 \supset S_1$	S_1 為 S_2 的真子集合	
$S_1 \subseteq S_2$, $S_2 \supseteq S_1$	S_1 為 S_2 的子集合	
$\cap_{i=1}^{n} S_i$	$S_1 \cap S_2 \cap \cdots \cap S_n$	
$\cup_{i=1}^{n} S_i$	$S_1 \cup S_2 \cup \cdots \cup S_n$	

區間

$(a,b)=\{x\,\vert\,a<x<b\}$	端點為 a, b 的開區間
$[a,b]=\{x\,\vert\,a\leqslant x\leqslant b\}$	端點為 a, b 的閉區間
$[a,b)=\{x\,\vert\,a\leqslant x<b\}$	
$(a,b]=\{x\,\vert\,a<x\leqslant b\}$	
$(a,\infty)=\{x\,\vert\,a<x\}$	
$[a,\infty)=\{x\,\vert\,a\leqslant x\}$	
$(-\infty,b)=\{x\,\vert\,x<b\}$	
$(-\infty,b]=\{x\,\vert\,x\leqslant b\}$	

不等式

性質. 令 $a,b,c\in\mathbb{R}$.

1. $a<b\implies a+c<b+c$
2. $a<b,\ c<d\implies a+c<b+d$
3. $a<b,\ c>0\implies ac<bc$
4. $a<b,\ c<0\implies ac>bc$
5. $0<a<b\implies \dfrac{1}{a}>\dfrac{1}{b}$

例. 解下列不等式.

1. $2x-3<x+4<3x-2$
2. $x^3>x$
3. $(2-x)(1-x)^2x^3\leqslant 0$
4. $-2<\dfrac{2x-3}{x+1}<1$

解.

1. $3<x<7$
2. $x^3-x>0\implies x(x^2-1)>0\implies x(x+1)(x-1)>0\implies x>1\vee -1<x<0$
3. $(2-x)(1-x)^2x^3\leqslant 0\implies (x-2)(x-1)^2x^3\geqslant 0\implies x\geqslant 2\vee x\leqslant 0\vee x=1$
4. $-2<\dfrac{2x-3}{x+1}<1\implies \left(-2<\dfrac{2x-3}{x+1}\right)\wedge\left(\dfrac{2x-3}{x+1}<1\right)\implies \left(\dfrac{4x-1}{x+1}>0\right)\wedge\left(\dfrac{x-4}{x+1}<0\right)$
   $\implies \left(x<-1\vee x>\dfrac{1}{4}\right)\wedge(-1<x<4)\implies \dfrac{1}{4}<x<4$

絕對值

令 $a\in\mathbb{R}$; a 的絕對值 (absolute value) $|a|$ 定義為 $|a|=\begin{cases}a & 若\ a\geqslant 0\\ -a & 若\ a<0\end{cases}$

性質. 若 $a>0$, 則

1. $|x|=a\iff x=\pm a$
2. $|x|<a\iff -a<x<a$
3. $|x|>a\iff x<-a\ \vee\ x>a$

性質. 若 $a,b\in\mathbb{R}$, 則

1. $|-a| = |a|$
2. $\sqrt{a^2} = |a|$
3. $|ab| = |a|\,|b|$
4. $\left|\dfrac{b}{a}\right| = \dfrac{|b|}{|a|}$
5. $|a+b| \leqslant |a| + |b|$
6. $\big|\,|a| - |b|\,\big| \leqslant |a-b|$

證 (不等式)**.**

- $(|a+b|)^2 = (a+b)^2 = a^2+2ab+b^2 = |a|^2+2ab+|b|^2 \leqslant |a|^2+2|ab|+|b|^2 = |a|^2+2|a|\,|b|+|b|^2 = (|a|+|b|)^2$, 故 $|a+b| \leqslant |a| + |b|$.
- $|a| = |(a-b)+b| \leqslant |a-b|+|b| \implies |a|-|b| \leqslant |a-b|$; $|b| = |(b-a)+a| \leqslant |b-a|+|a| = |a-b|+|a| \implies |a|-|b| \geqslant -|a-b|$. 故 $\big|\,|a|-|b|\,\big| \leqslant |a-b|$.

例. 解下列不等式與方程式.

1. $|5-2x| < 3$
2. $\left|\dfrac{2x-1}{x+1}\right| = 3$
3. $|x-1| - |x-10| \geqslant 5$

解.

1. $|5-2x| < 3 \implies -3 < 5-2x < 3 \implies -8 < -2x < -2 \implies 1 < x < 4$
2. $\left|\dfrac{2x-1}{x+1}\right| = 3 \implies \dfrac{2x-1}{x+1} = 3 \lor \dfrac{2x-1}{x+1} = -3 \implies x = -4 \lor x = -\dfrac{2}{5}$
3. 當 $x < 1$, $|x-1|-|x-10| \geqslant 5 \implies (1-x)-(10-x) \geqslant 5 \implies -9 \geqslant 5$, 不合. 當 $1 \leqslant x < 10$, $|x-1|-|x-10| \geqslant 5 \implies (x-1)-(10-x) \geqslant 5 \implies 2x \geqslant 16 \implies x \geqslant 8$, 則 $8 \leqslant x < 10$. 當 $x \geqslant 10$, $|x-1|-|x-10| \geqslant 5 \implies (x-1)-(x-10) \geqslant 5 \implies 9 \geqslant 5$ 恆成立. 綜上, $8 \leqslant x$.

平面幾何

性質.

- $P_1 : (x_1, y_1)$ 與 $P_2 : (x_2, y_2)$ 之距離為 $\sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$
- 一非鉛直線通過 $P_1 : (x_1, y_1)$ 與 $P_2 : (x_2, y_2)$, 則直線斜率 m 為 $m = \dfrac{\Delta y}{\Delta x} = \dfrac{y_2-y_1}{x_2-x_1}$
- 直線方程式表示法 — 點斜式: $y - y_1 = m(x-x_1)$; 斜截式: $y = mx+b$; 截距式: $\dfrac{x}{a} + \dfrac{y}{b} = 1$
- 兩非鉛直線 — 平行: 斜率相等; 垂直: 斜率相乘為 -1

0.2 函數

定義.

- 函數 (function) $f : A \to B$ 是一個對應關係: 對所有 $a \in A$, 存在唯一 $b \in B$, 使得 f 將 a 對應到 b. $\forall\, a \in A\ \exists!\, b \in B\ (f(a) = b)$.
- A — 定義域 (domain): $\operatorname{dom} f = A$; B — 對應域 (codomain): $\operatorname{codom} f = B$
  $f(A) = \{f(a) \mid a \in A\} \subseteq B$ — 值域 (range): $\operatorname{ran} f \equiv f(A)$

註 (函數形式)**.**

- 公式型 — $y = f(x) = 3x+1$, $\operatorname{dom} f = \operatorname{ran} f = \mathbb{R}$; $y = f(x) = 3x^2+1$, $\operatorname{dom} f = \mathbb{R}$, $\operatorname{ran} f = \{x \mid x \geqslant 1\}$; $y = f(x) = \sin \pi x$, $\operatorname{dom} f = \mathbb{R}$, $\operatorname{ran} f = [-1, 1]$.
- 抽象型 — 例: 令 $A = \{\text{Mon, Tue, Wed, Thu, Fri}\}$, $B = \{\text{a, b, c}, \ldots, \text{z}\}$, 定義函數 $f : A \to B$ 使得 f(工作日) = (開頭小寫英文字母): $f(\text{Mon}) = \text{m}$, $f(\text{Wed}) = \text{w}$, $f(\text{Tue}) = f(\text{Thu}) = \text{t}$, $f(\text{Fri}) = \text{f}$. 故 $\operatorname{dom} f = A$, $\operatorname{codom} f = B$, $\operatorname{ran} f = \{\text{m, w, t, f}\}$.

例. $f(x) = \sqrt{-x^2+x+2}$, 求 $\operatorname{dom} f$ 與 $\operatorname{ran} f$.

解. 由 $-x^2+x+2=-(x+1)(x-2)\geqslant 0 \implies (x+1)(x-2)\leqslant 0 \implies -1\leqslant x\leqslant 2$, $\operatorname{dom} f=[-1,2]$. 又 $-x^2+x+2=-\left(x-\frac{1}{2}\right)^2+\frac{9}{4}$, 當 $x\in[-1,2]$ 時, $f(x)\in\left[0,\sqrt{\frac{9}{4}}\right]=\left[0,\frac{3}{2}\right]$, 故 $\operatorname{ran} f=\left[0,\frac{3}{2}\right]$.

例. $f(x)=\dfrac{1}{(x-2)(x-3)}$, 求 $\operatorname{dom} f$ 與 $\operatorname{ran} f$.

解. $\operatorname{dom} f=\mathbb{R}\setminus\{2,3\}$. 令 $y=\dfrac{1}{(x-2)(x-3)}=\dfrac{1}{x^2-5x+6}\implies x^2-5x+\left(6-\dfrac{1}{y}\right)=0$. 當判別式 $\geqslant 0$ 時有實數解 $\implies (-5)^2-4\left(6-\dfrac{1}{y}\right)\geqslant 0\implies 1+\dfrac{4}{y}\geqslant 0\implies \dfrac{y+4}{y}\geqslant 0$, 故 $\operatorname{ran} f=\{y\mid y>0\lor y\leqslant -4\}$.

嵌射與蓋射

定義. 給定函數 $f:A\to B$.

- f 為嵌射 (one-to-one, injective): $\forall\, x_1,x_2\in A\ (f(x_1)=f(x_2)\implies x_1=x_2)$.
- f 為蓋射 (onto, surjective): $\forall\, b\in B\ \exists\, a\in A\ (f(a)=b)$.

例. 證明函數 $f:\mathbb{R}\to\mathbb{R}$, $f(x)=2x+1$ 為嵌射.

解. $\forall\, x_1,x_2\in\mathbb{R}$, $f(x_1)=f(x_2)\implies (2x_1+1)=(2x_2+1)\implies x_1=x_2$.

例.

- 說明函數 $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^2$ 不為嵌射.
- 證明函數 $f:\mathbb{R}_+=\{x\mid x\geqslant 0\}\to\mathbb{R}$, $f(x)=x^2$ 為嵌射.

解.

- 取 $x_1=1$, $x_2=-1$, $x_1\neq x_2$, 但 $f(x_1)=f(x_2)=1$.
- $\forall\, x_1,x_2\in\mathbb{R}$, $f(x_1)=f(x_2)\implies x_1^2=x_2^2\implies (x_1-x_2)(x_1+x_2)=0\implies x_1-x_2=0\lor x_1+x_2=0\implies x_1=x_2\lor x_1=x_2=0\implies x_1=x_2$.

例. 證明函數 $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^3$ 為嵌射.

解. $\forall\, x_1,x_2\in\mathbb{R}$, $f(x_1)=f(x_2)\implies x_1^3=x_2^3\implies (x_1-x_2)(x_1^2+x_1x_2+x_2^2)=0\implies x_1-x_2=0\lor x_1^2+x_1x_2+x_2^2=0\implies x_1=x_2\lor\left(x_1+\dfrac{x_2}{2}\right)^2+\dfrac{3x_2^2}{4}=0\implies x_1=x_2\lor\left(x_2=0\land x_1+\dfrac{x_2}{2}=0\right)\implies x_1=x_2\lor x_2=x_1=0\implies x_1=x_2$.

0.3 函數運算

$(f\pm g)(x)=f(x)\pm g(x)$	$\operatorname{dom}(f\pm g)=\operatorname{dom} f\cap\operatorname{dom} g$
$(f\cdot g)(x)=f(x)\cdot g(x)$	$\operatorname{dom}(f\cdot g)=\operatorname{dom} f\cap\operatorname{dom} g$
$\left(\dfrac{f}{g}\right)(x)=\dfrac{f(x)}{g(x)}$	$\operatorname{dom}\dfrac{f}{g}=\operatorname{dom} f\cap\operatorname{dom} g\setminus\{x\mid g(x)=0\}$
$(f\circ g)(x)=f(g(x))$	$\operatorname{dom}(f\circ g)=\{x\in\operatorname{dom} g\mid g(x)\in\operatorname{dom} f\}$

例.

1. 設 $f\left(x-\dfrac{1}{x}\right)=x^2+\dfrac{1}{x^2}$, 求 $f(2)$.

2. 設 $f(x)=x$, $g(x)=\dfrac{1}{x}$, $h(x)=(f\cdot g)(x)=x\cdot\dfrac{1}{x}=1$, 求 $\operatorname{dom} h$.
3. 設 $F(x)=\sin^3(x+3)$, 求函數 f, g, h 使得 $F=f\circ g\circ h$.
4. 設 $f(x)=\dfrac{x+1}{1+\frac{1}{x+1}}$, 求 $\operatorname{dom} f$.
5. 設 $f(x)=\sqrt{x}$, $g(x)=\sqrt{2-x}$, 求 $f\circ f$, $f\circ g$, $g\circ f$, $g\circ g$, 以及其定義域.
6. 設 $g(x)=2x+1$, $h(x)=4x^2+4x+7$.
   - 求函數 $f(x)$ 使得 $f\circ g=h$.
   - 求函數 $f(x)$ 使得 $g\circ f=h$.
7. 若 $f_0(x)=\dfrac{x}{x+1}$, $f_{n+1}=f_0\circ f_n$, $n=0,1,2,\ldots$. 求 $f_n(x)$ 之公式.

解.

1. 若 $x-\dfrac{1}{x}=2$, $4=\left(x-\dfrac{1}{x}\right)^2=x^2-2+\dfrac{1}{x^2}\implies x^2+\dfrac{1}{x^2}=6$. 由 $f\left(x-\dfrac{1}{x}\right)=x^2+\dfrac{1}{x^2}$, $f(2)=6$.
2. 由 $\operatorname{dom}(f\cdot g)=\operatorname{dom} f\cap\operatorname{dom} g$, $\operatorname{dom} h=\mathbb{R}\cap\big(\mathbb{R}\setminus\{0\}\big)=\mathbb{R}\setminus\{0\}$.
3. $f(x)=x^3$, $g(x)=\sin x$, $h(x)=x+3$.
4. 由 $\operatorname{dom}\dfrac{f}{g}=\operatorname{dom} f\cap\operatorname{dom} g\setminus\{x\,|\,g(x)=0\}$, $\operatorname{dom} f=\mathbb{R}\cap\big(\mathbb{R}\cap\big(\mathbb{R}\setminus\{-1\}\big)\big)\setminus\big\{x\,|\,1+\frac{1}{x+1}=0\big\}$ $=\mathbb{R}\setminus\{-1,-2\}$
5. $f\circ f=\sqrt[4]{x}$, $f\circ g=\sqrt[4]{2-x}$, $g\circ f=\sqrt{2-\sqrt{x}}$, $g\circ g=\sqrt{2-\sqrt{2-x}}$, $\operatorname{dom} f\circ f=[0,\infty)$, $\operatorname{dom} f\circ g=(-\infty,2]$, $\operatorname{dom} g\circ f=[0,4]$, $\operatorname{dom} g\circ g=[-2,2]$.
6. $f(x)=x^2+6$, $f(x)=2x^2+2x+3$.
7. 使用數學歸納法驗證 $f_n(x)=\dfrac{x}{(n+1)\,x+1}$.

0.4 函數圖形

定義. 若 A, $B\subseteq\mathbb{R}$, 則函數 $f:A\to B$ 稱為實數值函數 (real-valued function) , 集合 $\{(x,f(x))\,|\,x\in A\}$ 稱為 f 的圖形 (graph) .

性質. 函數 / 圖形判斷法

- 垂直線判斷法: 函數圖形 $\iff$ 任一垂直線與其至多交於一點
- 水平線判斷法: 嵌射圖形 $\iff$ 任一水平線與其至多交於一點

性質. 變換後圖形方程式

- 垂直平移: $y=f(x)+h$
- 水平平移: $y=f(x+h)$
- 垂直伸縮: $y=c\,f(x)$
- 水平伸縮: $y=f(c\,x)$
- $y=-f(x)$ 為 $y=f(x)$ 對 x 軸的鏡射.
- $y=f(-x)$ 為 $y=f(x)$ 對 y 軸的鏡射.

0.5 函數範例

例. 分段定義函數

- $|x|=\begin{cases}x & x\geqslant 0\\ -x & x<0\end{cases}$

- (最大整數 / Gauss / 地板) 函數 (greatest integer / Gauss / floor) $\lfloor x \rfloor$
  $\lfloor x \rfloor = n$, 若 $n \leqslant x < n+1$, $n \in \mathbb{Z}$. $\lfloor x \rfloor$ 為小於或等於 x 的最大整數.
- 天花板函數 (ceiling) $\lceil x \rceil$
  $\lceil x \rceil = n+1$, 若 $n < x \leqslant n+1$, $n \in \mathbb{Z}$. $\lceil x \rceil$ 為大於或等於 x 的最小整數.
- $\lceil x \rceil = -\lfloor -x \rfloor$.
- 若 g, h 為實數值函數, 則 $f(x) = \max\{g(x), h(x)\} = \dfrac{|g(x)-h(x)|}{2} + \dfrac{g(x)+h(x)}{2}$.

0.6 函數特性

奇偶性

定義. 給定實數值函數 f, $\forall x \in \operatorname{dom} f$

- 若 $f(-x) = f(x)$, 則 f 為偶函數 (even function) .
- 若 $f(-x) = -f(x)$, 則 f 為奇函數 (odd function) .

性質. 任意實數值函數可唯一表示成一個偶函數與一個奇函數的和.

證. 設實數值函數為 f.

- (存在性) 令 $E(x) = \dfrac{f(x)+f(-x)}{2}$, $O(x) = \dfrac{f(x)-f(-x)}{2}$, 則 E 為偶函數, O 為奇函數, $f(x) = E(x)+O(x)$.
- (唯一性) 已知 $f(x) = E_1(x)+O_1(x) = E_2(x)+O_2(x)$. 令 $E_1(x)-E_2(x) = e(x)$, $O_1(x)-O_2(x) = o(x)$, 則 $e(x)$ 為偶函數, $o(x)$ 為奇函數, 且 $e(x)+o(x) = 0$. 若 $x \leftarrow -x$, 則 $e(-x)+o(-x) = 0 \implies e(x)-o(x) = 0$, 故 $e(x) = 0$, $o(x) = 0$.

增減性

定義. 給定實數值函數 f 與區間 I. $\forall x, y \in I$, $x < y$:

- 若 $f(x) < f(y)$, 則稱 f 在 I 為嚴格遞增 / 嚴格上升 (increasing) .
- 若 $f(x) > f(y)$, 則稱 f 在 I 為嚴格遞減 / 嚴格下降 (decreasing) .
- 若 $f(x) \leqslant f(y)$, 則稱 f 在 I 為遞增 / 上升 (non-decreasing) .
- 若 $f(x) \geqslant f(y)$, 則稱 f 在 I 為遞減 / 下降 (non-increasing) .

0.7 反函數

定義. 若函數 f 為嵌射, 則其反函數 $f^{-1} : \operatorname{ran} f \to \operatorname{dom} f$ 定義為 $f^{-1}(b) = a \iff f(a) = b$, 其中 $a \in \operatorname{dom} f$, $b \in \operatorname{ran} f$.

性質 (常用規則).

1. $f^{-1}(y) = x \iff f(x) = y$
2. $\operatorname{dom} f^{-1} = \operatorname{ran} f$, $\operatorname{ran} f^{-1} = \operatorname{dom} f$
3. $f^{-1}(x) = (f(x))^{-1} \neq \dfrac{1}{f(x)}$
4. $(f^{-1} \circ f)(x) = x$, $\forall x \in \operatorname{dom} f$
5. $(f \circ f^{-1})(y) = y$, $\forall y \in \operatorname{dom} f^{-1} = \operatorname{ran} f$
6. $y = f(x)$ 與 $y = f^{-1}(x)$ 之圖形對 $y = x$ 對稱.
7. 若 f 為嚴格遞增或嚴格遞減函數, 則 f 為嵌射 $\implies$ 存在 f^{-1}.

例. 令 $f(x) = \sin x$, 若定義在

- $[0, \pi]$ 時, 非為嵌射 (一對一)
- $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$ 時為嚴格遞增, 存在反函數 $\sin^{-1} : [-1, 1] \to \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

例.

1. 求 $f(x) = x^3 + 2$ 的反函數.
2. 求 $f(x) = x^2$, $x \geqslant 0$ 的反函數, 並求 $x \leqslant 0$ 時的反函數.
3. 求 $f(x) = \dfrac{1+9x}{4-x}$, $x < 4$ 的反函數.

解.

1. $y = x^3 + 2$; $x \longleftrightarrow y$: $x = y^3 + 2 \implies y^3 = x - 2 \implies y = \sqrt[3]{x-2} \implies f^{-1}(x) = \sqrt[3]{x-2}$.
2. $y = x^2$, $x \geqslant 0$; $x \longleftrightarrow y$: $x = y^2 \implies y^2 = x \implies y = \sqrt{x} \implies f^{-1}(x) = \sqrt{x}$; $f^{-1} : [0, \infty) \to [0, \infty)$.
   $y = x^2$, $x \leqslant 0$; $x \longleftrightarrow y$: $x = y^2 \implies y^2 = x \implies y = -\sqrt{x} \implies f^{-1}(x) = -\sqrt{x}$; $f^{-1} : [0, \infty) \to (-\infty, 0]$.
3. $f(x) = \dfrac{1+9x}{4-x} \implies y = \dfrac{1+9x}{4-x}$; $x \longleftrightarrow y$: $x = \dfrac{1+9y}{4-y} \implies y = \dfrac{4x-1}{x+9} \implies f^{-1}(x) = \dfrac{4x-1}{x+9}$.
   檢驗: $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = \dfrac{4 \cdot \frac{1+9x}{4-x} - 1}{\frac{1+9x}{4-x} + 9} = \dfrac{\frac{4+36x-4+x}{4-x}}{\frac{1+9x+36-9x}{4-x}} = \dfrac{37x}{37} = x$; $(f \circ f^{-1})(x) = f(f^{-1}(x)) = \dfrac{1 + 9 \cdot \frac{4x-1}{x+9}}{4 - \frac{4x-1}{x+9}} = \dfrac{\frac{x+9+36x-9}{x+9}}{\frac{4x+36-4x+1}{x+9}} = \dfrac{37x}{37} = x$.

0.8 指數函數

$y = f(x) = a^x$, $a > 0 \wedge a \neq 1$.

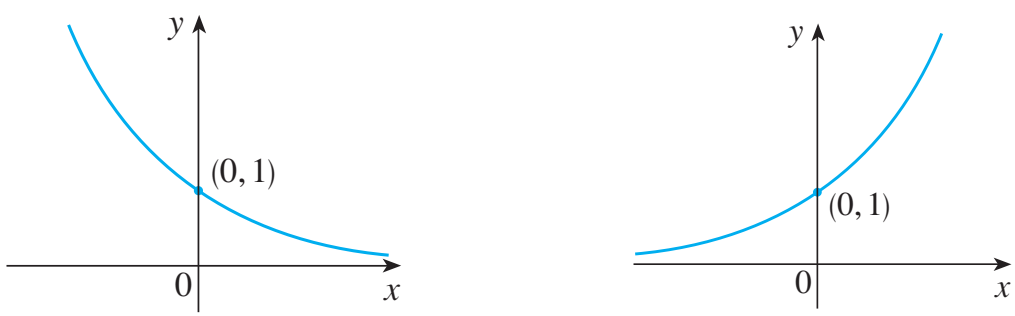


圖 1: $y = a^x$: 圖左 $0 < a < 1$, 圖右 $a > 1$

性質. 若 a, $b > 0$, x, $y \in \mathbb{R}$, 則

- $a^x \cdot a^y = a^{x+y}$
- $a^{-x} = \dfrac{1}{a^x}$
- $\dfrac{a^x}{a^y} = a^{x-y}$
- $(a^x)^y = a^{xy} = (a^y)^x$
- $a^x \cdot b^x = (ab)^x$
- $\dfrac{a^x}{b^x} = \left(\dfrac{a}{b}\right)^x$

0.9 對數函數

$y = f(x) = \log_a x$, $a > 0 \wedge a \neq 1$.

性質. 給定 $a > 0 \wedge a \neq 1$, $x > 0$, $y \in \mathbb{R}$.

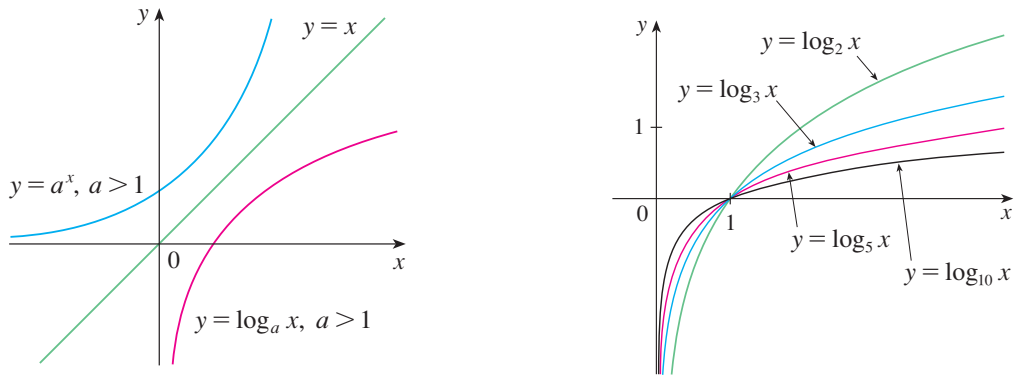


圖 2: $y = \log_a x$

- $\log_a x = y \iff a^y = x$
- $\log_a a^y = y$
- $a^{\log_a x} = x$

性質. 給定 $b > 0$, $x > 0$, $a > 0 \wedge a \neq 1$, $c > 0 \wedge c \neq 1$, $r \in \mathbb{R}$.

- $\log_a bx = \log_a b + \log_a x$
- $\log_a x^r = r \log_a x$
- $\log_a x = \dfrac{\log_c x}{\log_c a}$

例. 解下列 x 的方程式與不等式.

1. $\log_{10} x + \log_{10}(x - 21) = 2$
2. $\log_2 \left(x^2 - 2x - 2\right) \leqslant 0$
3. $x^{\log_3 x} = 27x^2$
4. $3^{\log_3 7} - 4^{\log_4 2} = 5^{\log_5 x - \log_5 x^2}$
5. $\left(\dfrac{4}{3}\right)^{-x^2+\frac{3}{2}x+1} < \left(\dfrac{\sqrt{3}}{2}\right)^{-4x+1}$

解.

1. $\log_{10}(x^2 - 21x) = \log_{10} 10^2 \implies x^2 - 21x - 100 = 0 \implies (x-25)(x+4) = 0 \implies x = 25 \vee x = -4$ (不合).
2. $x^2 - 2x - 2 > 0 \wedge x^2 - 2x - 2 \leqslant 1 \implies x \in [-1, 1-\sqrt{3}) \cup (1+\sqrt{3}, 3]$.
3. 方程式兩邊取 $\log_3$ 得 $\log_3\left(x^{\log_3 x}\right) = \log_3(27x^2) \implies (\log_3 x)^2 = \log_3 27 + 2\log_3 x$. 令 $\log_3 x = y$, 則 $y^2 = 3 + 2y \implies y = 3 \vee -1 \implies x = 27 \vee -\dfrac{1}{3}$.
4. $7 - 2 = \dfrac{1}{x} \implies x = \dfrac{1}{5}$.
5. $\left(\dfrac{4}{3}\right)^{-x^2+\frac{3}{2}x+1} < \left(\dfrac{\sqrt{3}}{2}\right)^{-4x+1} \implies \left(\dfrac{2}{\sqrt{3}}\right)^{-2x^2-x+3} < 1 \implies -2x^2 - x + 3 < 0 \implies x > 1 \vee x < -\dfrac{3}{2}$.

例. 若 $x > 0$, 若 $f(x) = (32\,x)^{7-\log_2 x}$ 在 $x = a$ 有最大值 M, 求 (a, M).

解. 令 $u = \log_2 x$, 則 $x = 2^u$; $f(u) = (32 \cdot 2^u)^{7-u} = (2^5 \cdot 2^u)^{7-u} = 2^{(5+u)(7-u)} = 2^{-u^2+2u+35} = 2^{-(u-1)^2+36}$. 故 $f(u)$ 在 $u = 1$, 亦即 $x = 2$ 有最大值 2^{36}; $(a, M) = (2, 2^{36})$.

例. 證明 $f(x) = \log_2\left(x + \sqrt{x^2+1}\right)$ 為奇函數, 並求其反函數.

解.

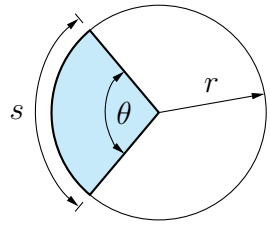
- $f(-x) = \log_2\left(-x + \sqrt{x^2+1}\right) = \log_2\left(\dfrac{\left(\sqrt{x^2+1}\right)^2 - x^2}{\sqrt{x^2+1} + x}\right) = \log_2\left(\dfrac{1}{x + \sqrt{x^2+1}}\right) = -f(x)$.
- $y = \log_2\left(x + \sqrt{x^2+1}\right)$; $x \longleftrightarrow y$: $x = \log_2\left(y + \sqrt{y^2+1}\right) \implies 2^x - y = \sqrt{y^2+1} \implies 2^{2x} - 2 \cdot 2^x y + y^2 = y^2 + 1 \implies y = \dfrac{2^x - 2^{-x}}{2}$.

0.10 三角函數

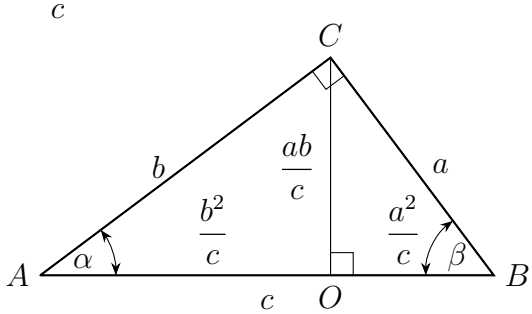
定義.

- 弧度 / 弳度 (radian)

$$\theta = \frac{s}{r}$$

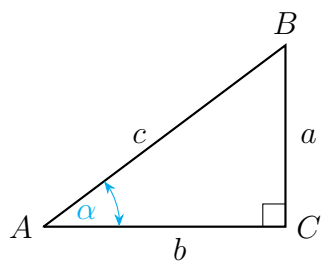


- 直角三角形: 畢氏定理 (Pythagorean Theorem) $c^2 = a^2 + b^2$
  證明: 由直角三角形面積公式 $\frac{1}{2}c \cdot \overline{CO} = \frac{1}{2}a \cdot b \implies \overline{CO} = \frac{ab}{c}$. 又 $\Delta ABC \sim \Delta CBO \sim \Delta ACO$, 則 $\overline{AC} : \overline{BC} = \overline{CO} : \overline{BO} = \overline{AO} : \overline{CO} \implies b : a = \frac{ab}{c} : \overline{BO} = \overline{AO} : \frac{ab}{c} \implies \overline{BO} = \frac{a^2}{c}, \overline{AO} = \frac{b^2}{c}$. 由 $\overline{AB} = \overline{AO} + \overline{BO} \implies c = \frac{b^2}{c} + \frac{a^2}{c} \implies c^2 = a^2 + b^2$.



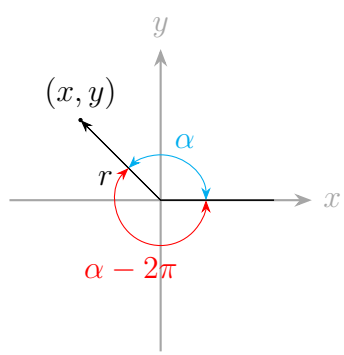
- 銳角三角函數

$$\sin\alpha = \frac{a}{c} \qquad \csc\alpha = \frac{c}{a}$$
$$\cos\alpha = \frac{b}{c} \qquad \sec\alpha = \frac{c}{b}$$
$$\tan\alpha = \frac{a}{b} \qquad \cot\alpha = \frac{b}{a}$$



- 廣義角三角函數

$$\sin\alpha = \frac{y}{r} \qquad \csc\alpha = \frac{r}{y}$$
$$\cos\alpha = \frac{x}{r} \qquad \sec\alpha = \frac{r}{x}$$
$$\tan\alpha = \frac{y}{x} \qquad \cot\alpha = \frac{x}{y}$$



性質. 常用規則

- $\sin(-\alpha) = -\sin\alpha$, $\cos(-\alpha) = \cos\alpha$.
- $\forall\, n \in \mathbb{Z}$: $\sin n\pi = 0$, $\cos n\pi = (-1)^n$.
- $\sin^2\alpha + \cos^2\alpha = 1$, $\tan^2\alpha + 1 = \sec^2\alpha$, $\cot^2\alpha + 1 = \csc^2\alpha$.

性質. 和角公式 (addition formula)

- $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
- $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$

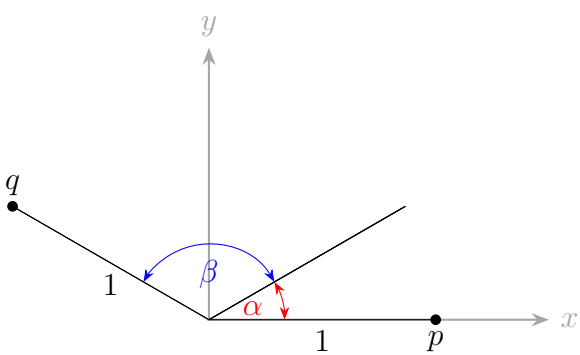
性質. 兩倍角公式

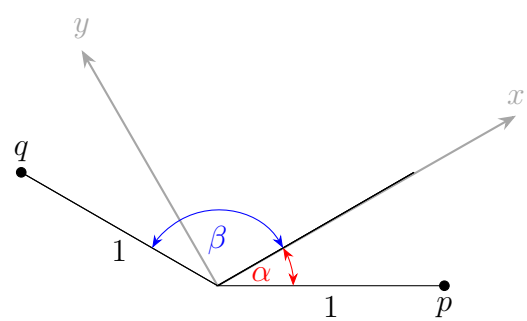
- $\sin 2\alpha = 2\sin\alpha\cos\alpha$
- $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$
- $\cos^2 x = \dfrac{1+\cos 2x}{2}$, $\sin^2 x = \dfrac{1-\cos 2x}{2}$

性質. 積化和差公式

- $\sin\alpha\cos\beta = \dfrac{1}{2}\big(\sin(\alpha-\beta) + \sin(\alpha+\beta)\big)$
- $\sin\alpha\sin\beta = \dfrac{1}{2}\big(\cos(\alpha-\beta) - \cos(\alpha+\beta)\big)$
- $\cos\alpha\cos\beta = \dfrac{1}{2}\big(\cos(\alpha-\beta) + \cos(\alpha+\beta)\big)$

證 (和角公式)**.** 設 $p = (1,0)$, $q = (\cos(\alpha+\beta), \sin(\alpha+\beta))$. 座標系旋轉 α 後, $p = (\cos\alpha, -\sin\alpha)$, $q = (\cos\beta, \sin\beta)$. p, q 距離不變 $\implies \sqrt{(\cos(\alpha+\beta)-1)^2 + \sin^2(\alpha+\beta)} = \sqrt{(\cos\beta - \cos\alpha)^2 + (\sin\beta + \sin\alpha)^2}$ $\implies \cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$. 代入 $\alpha = x$, $\beta = -\dfrac{\pi}{2}$ 可得 $\sin x = \cos\big(x - \dfrac{\pi}{2}\big)$, 則 $\sin(\alpha+\beta) = \cos\big(\alpha + \beta - \dfrac{\pi}{2}\big) = \cos\alpha\cos\big(\beta - \dfrac{\pi}{2}\big) - \sin\alpha\sin\big(\beta - \dfrac{\pi}{2}\big) = \cos\alpha\sin\beta - \sin\alpha\cos\big(\beta - \pi\big) = \cos\alpha\sin\beta - \sin\alpha(\cos\beta\cos\pi + \sin\beta\sin\pi) = \cos\alpha\sin\beta + \sin\alpha\cos\beta$.





0.11 反三角函數

定義. 在以下定義域上之三角函數為嵌射:

$$\sin x : \Big[-\frac{\pi}{2}, \frac{\pi}{2}\Big] \to [-1,1] \qquad \csc x : \Big(0, \frac{\pi}{2}\Big] \cup \Big(\pi, \frac{3\pi}{2}\Big] \to (-\infty,-1] \cup [1,\infty)$$

$$\cos x : \big[0, \pi\big] \to [-1,1] \qquad \sec x : \Big[0, \frac{\pi}{2}\Big) \cup \Big[\pi, \frac{3\pi}{2}\Big) \to (-\infty,-1] \cup [1,\infty)$$

$$\tan x : \Big(-\frac{\pi}{2}, \frac{\pi}{2}\Big) \to (-\infty,\infty) \qquad \cot x : \big(0, \pi\big) \to (-\infty,\infty)$$

故存在反三角函數:

$$\sin^{-1} x : [-1,1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \qquad \csc^{-1} x : (-\infty,-1] \cup [1,\infty) \to \left(0, \frac{\pi}{2}\right] \cup \left(\pi, \frac{3\pi}{2}\right]$$

$$\cos^{-1} x : [-1,1] \to [0,\pi] \qquad \sec^{-1} x : (-\infty,-1] \cup [1,\infty) \to \left[0, \frac{\pi}{2}\right) \cup \left[\pi, \frac{3\pi}{2}\right)$$

$$\tan^{-1} x : (-\infty,\infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \qquad \cot x : (-\infty,\infty) \to (0,\pi)$$

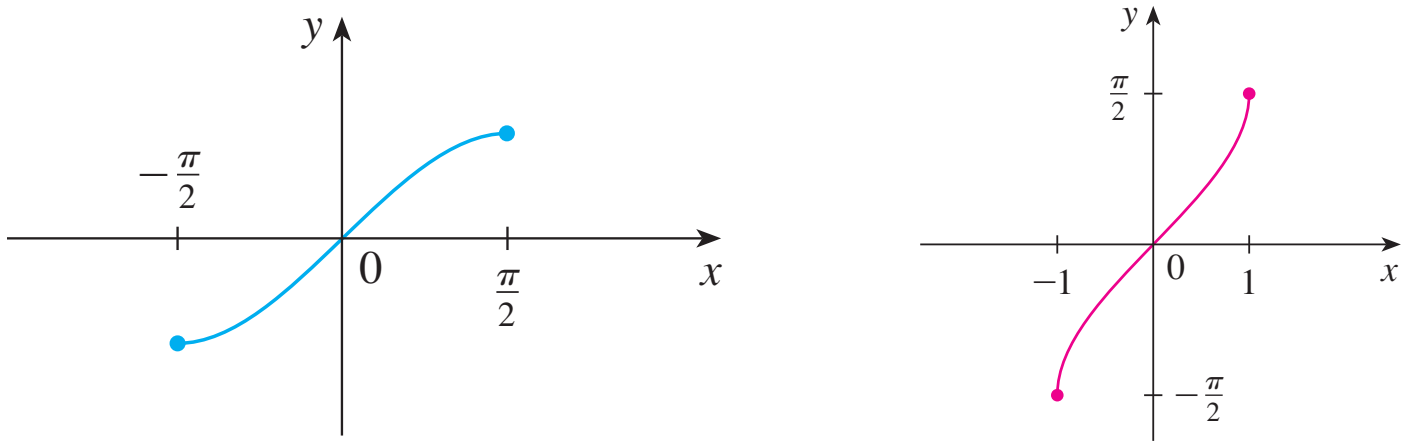


圖 3: $y = \sin x$, $y = \sin^{-1} x$

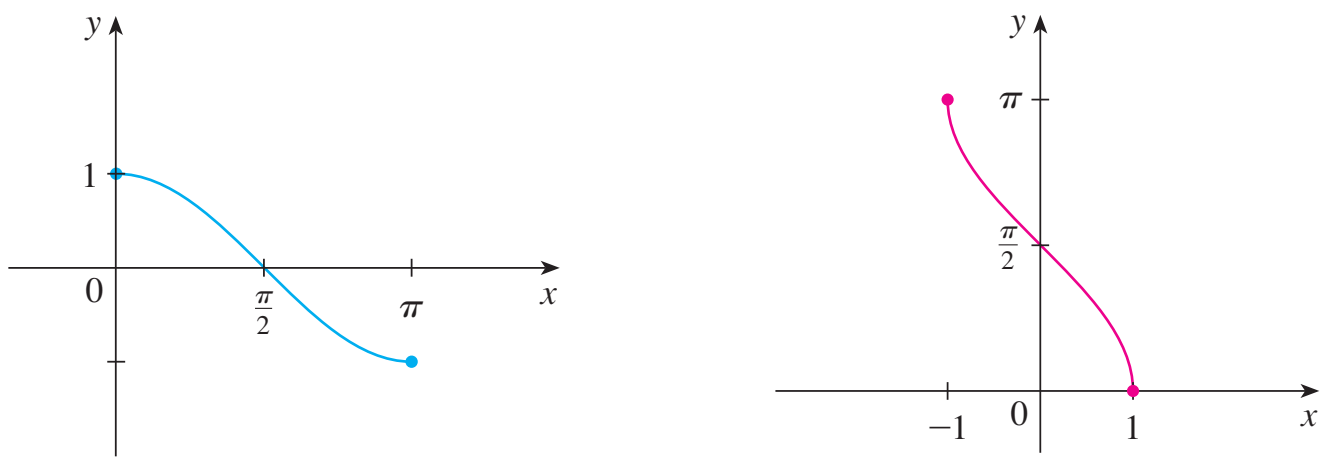


圖 4: $y = \cos x$, $y = \cos^{-1} x$

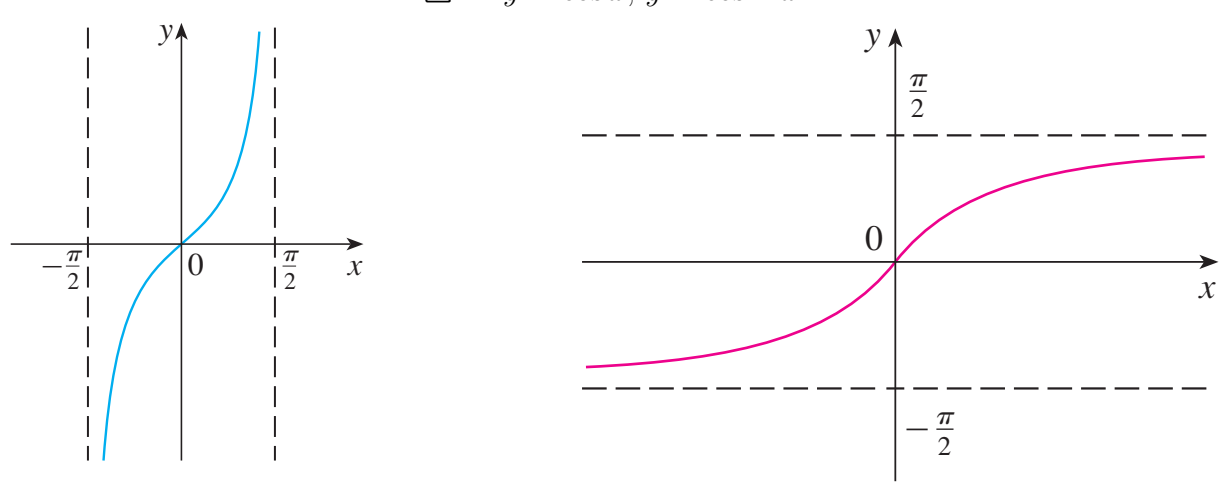


圖 5: $y = \tan x$, $y = \tan^{-1} x$

性質.

- 當 $-\dfrac{\pi}{2} \leqslant x \leqslant \dfrac{\pi}{2}$, $\sin^{-1}(\sin x) = x$, $\tan^{-1}(\tan x) = x$; 當 $0 \leqslant x \leqslant \pi$, $\cos^{-1}(\cos x) = x$.
- 當 $-1 \leqslant x \leqslant 1$, $\sin(\sin^{-1} x) = x$, $\cos(\cos^{-1} x) = x$; 當 $-\infty < x < \infty$, $\tan(\tan^{-1} x) = x$.
- $\sin^{-1}(-x) = -\sin^{-1} x$; $\tan^{-1}(-x) = -\tan^{-1} x$; $\cos^{-1}(-x) = \pi - \cos^{-1} x$.

例. 基本反三角函數求值.

1. $\sin^{-1} \dfrac{\sqrt{3}}{2} = \dfrac{\pi}{3}$
2. $\cos^{-1}\left(-\dfrac{1}{2}\right) = \dfrac{2\pi}{3}$
3. $\tan^{-1} 1 = \dfrac{\pi}{4}$
4. $\cos\left(\sin^{-1} \dfrac{3}{5}\right) = \dfrac{4}{5}$

例. 若 $\alpha = \sin^{-1} \dfrac{2}{3}$, 求 $\cos\alpha$, $\tan\alpha$, $\cot\alpha$, $\sec\alpha$, $\csc\alpha$.

解. $\cos\alpha = \dfrac{\sqrt{5}}{3}$, $\tan\alpha = \dfrac{2}{\sqrt{5}}$, $\cot\alpha = \dfrac{\sqrt{5}}{2}$, $\sec\alpha = \dfrac{3}{\sqrt{5}}$, $\csc\alpha = \dfrac{3}{2}$.

例. 若 $f(x) = \dfrac{1}{\sin^{-1}\left(\frac{1}{x}\right)}$, 求 $\operatorname{dom} f$.

解. $\operatorname{dom} f = \left\{x \mid -1 \leqslant \frac{1}{x} \leqslant 1\right\} = (-\infty, -1] \cup [1, \infty)$.

例. 令 $f(x) = \sin(\sin^{-1} x)$, $g(x) = \sin^{-1}(\sin x)$, 求 $\operatorname{dom} f$, $\operatorname{dom} g$ 與其圖形.

解.

- $\operatorname{dom} f = \operatorname{dom} \sin^{-1} = [-1, 1]$; $f(x) = x$, $\forall x \in [-1, 1]$.
- $\operatorname{dom} g = \mathbb{R}$; $g(x) = \begin{cases} -x + (2n+1)\pi & x \in [(2n+\frac{1}{2})\pi,\ (2n+\frac{3}{2})\pi) \\ x - (2n+2)\pi & x \in [(2n+\frac{3}{2})\pi,\ (2n+\frac{5}{2})\pi) \end{cases}$ $\quad n \in \mathbb{Z}$.

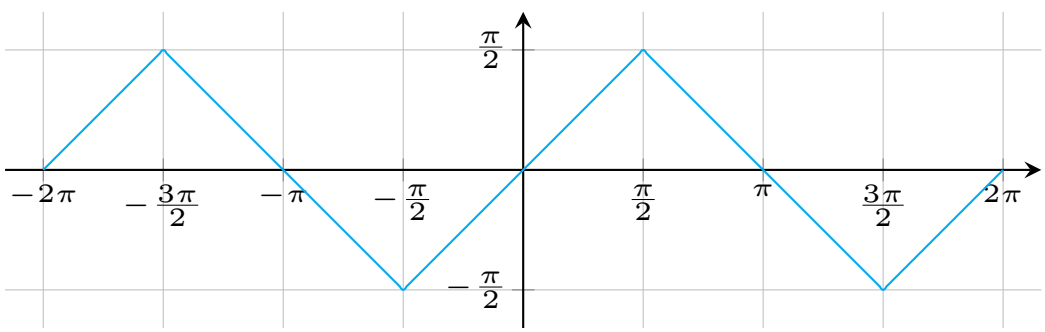


圖 6: $g(x) = \sin^{-1}(\sin x)$

例. 將 $\sin(\cos^{-1} x)$ 與 $\tan(\cos^{-1} x)$ 化簡為 x 的 (不含三角函數之) 表示式, 其中 $-1 \leqslant x \leqslant 1$.

解. 令 $u = \cos^{-1} x$, 則 $0 \leqslant u \leqslant \pi$, $\cos u = x$, $\sin u = +\sqrt{1-x^2}$, $\tan u = \dfrac{\sqrt{1-x^2}}{x}$.

例. 化簡以下表示式.

1. $\cos(\sin^{-1} x)$
2. $\sin(\tan^{-1} x)$
3. $\sin(2\tan^{-1} x)$

解.

1. $\cos(\sin^{-1} x) = \sqrt{1-x^2}$
2. $\sin(\tan^{-1} x) = \dfrac{x}{\sqrt{1+x^2}}$
3. $\sin(2\tan^{-1} x) = \dfrac{2x}{1+x^2}$

例. 將 $\sin(\cos^{-1} x + \tan^{-1} y)$ 化簡為 x, y 的 (不含三角函數之) 表示式, 其中 $-1 \leqslant x \leqslant 1$, $y \in \mathbb{R}$.

解. 令 $u = \cos^{-1} x$, $v = \tan^{-1} y$, 則 $\cos u = x$, $\tan v = y$, 由此 $\sin u = \sqrt{1-x^2}$, $\cos v = \dfrac{1}{\sqrt{1+y^2}}$, $\sin v = \dfrac{y}{\sqrt{1+y^2}}$; 原式為 $\sin(\cos^{-1} x + \tan^{-1} y) = \sin(u+v) = \sin u \cos v + \cos u \sin v = \sqrt{1-x^2} \cdot \dfrac{1}{\sqrt{1+y^2}} + x \cdot \dfrac{y}{\sqrt{1+y^2}}$.

例. 證明 $\cos^{-1}(1-2x^2) = 2\sin^{-1} x$, $0 \leqslant x \leqslant 1$.

解. 令 $u = \sin^{-1} x$, 則 $\cos^{-1}(1-2x^2) = \cos^{-1}(1-2\sin^2 u) = \cos^{-1}(\cos 2u) = 2u = 2\sin^{-1} x$.

例. 解 $2\sin^{-1} x + \cos^{-1} x = \pi$.

解. 令 $\sin^{-1} x = u$, $\cos^{-1} x = v$, 則 $\sin u = \cos v = x$, $\cos u = \sin v = \sqrt{1-x^2}$. 方程式兩邊取 cos $\implies \cos(2u+v) = -1 \implies \cos 2u \cos v - \sin 2u \sin v = (1-2\sin^2 u)\cos v - 2\sin u \cos u \sin v = -1 \implies (1-2x^2)x - 2x(1-x^2) = -1 \implies x = 1$.